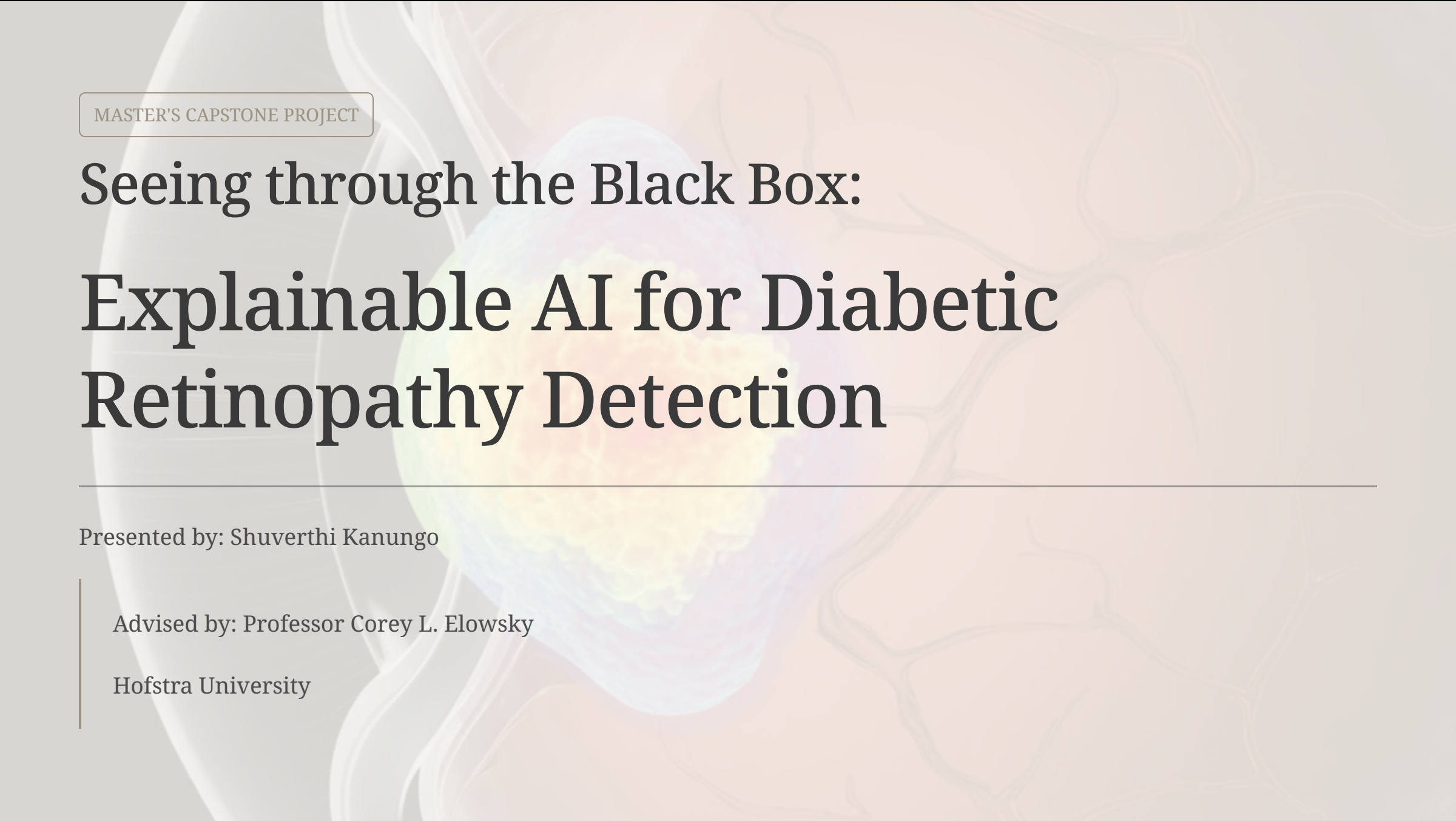




MASTER'S CAPSTONE PROJECT

Seeing through the Black Box: Explainable AI for Diabetic Retinopathy Detection

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THE PROBLEM

Diabetic Retinopathy: A Silent Crisis

460M+

People with Diabetes
Globally

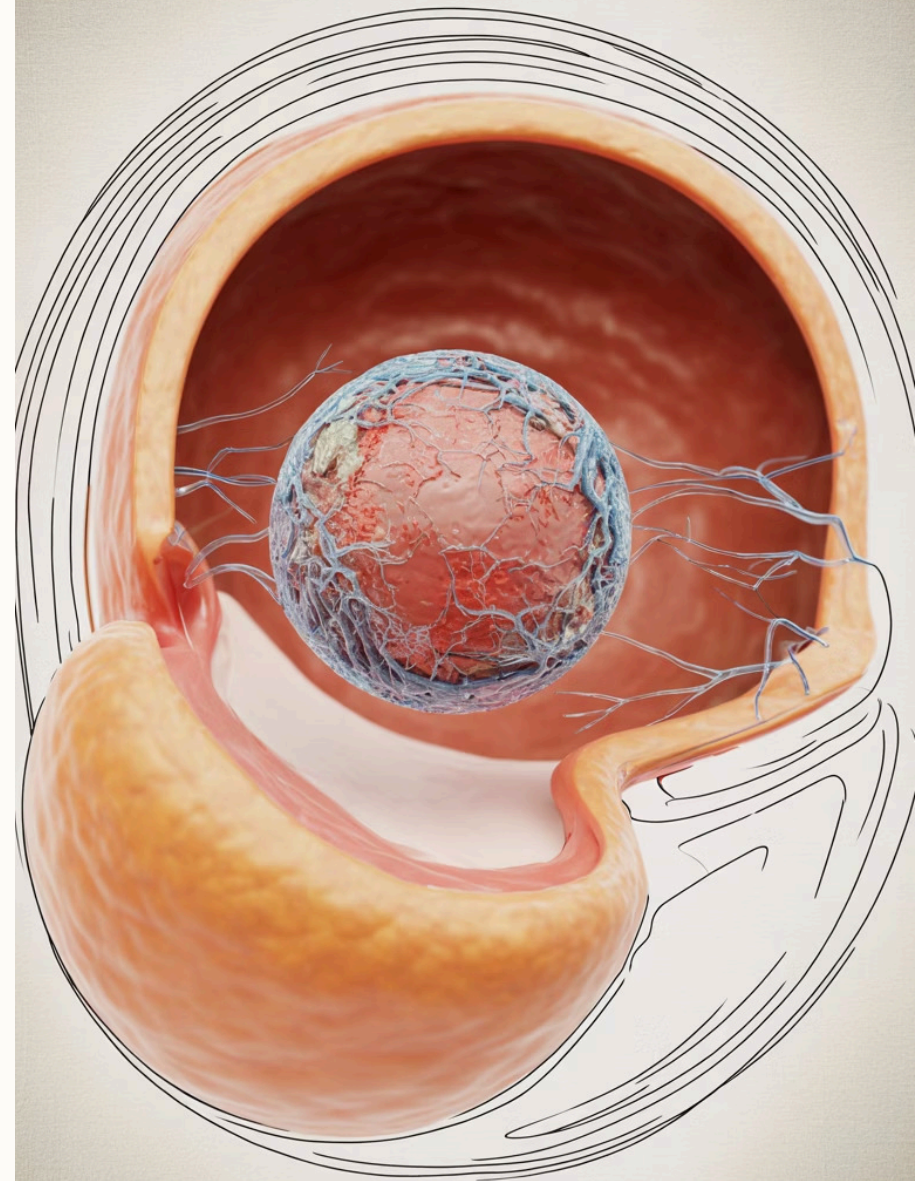
#1

Cause of Blindness
Preventable blindness worldwide

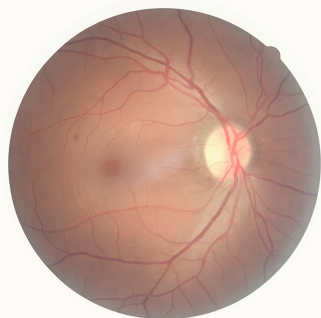
0

Early Symptoms
Until it is too late

- Diabetes is one of the fastest-growing diseases globally.
- High blood sugar damages blood vessels in the eye.
- Without early detection → permanent blindness.



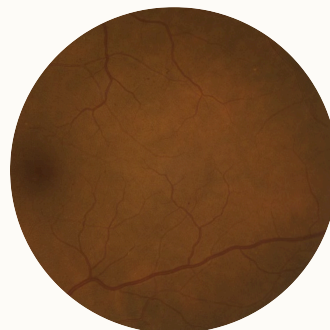
DR Severity Scale



Grade 0

No DR

No retinal damage



Grade 1

Mild

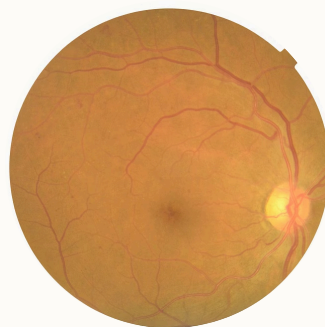
Tiny microaneurysms



Grade 2

Moderate

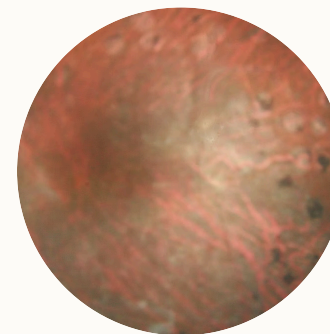
Hard exudates, small bleeds



Grade 3

Severe

Widespread hemorrhages



Grade 4

Proliferative

New abnormal blood vessels

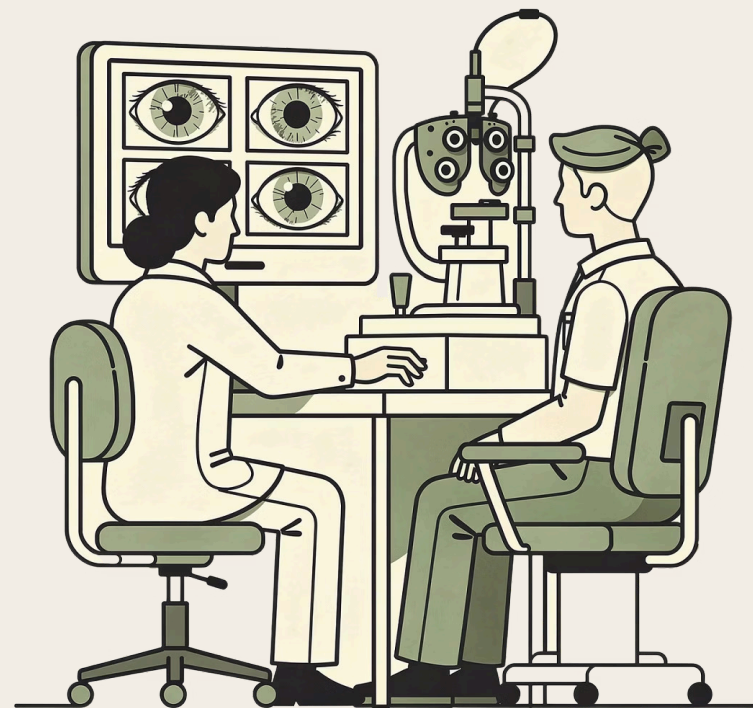
THE PROBLEM

The Screening Gap

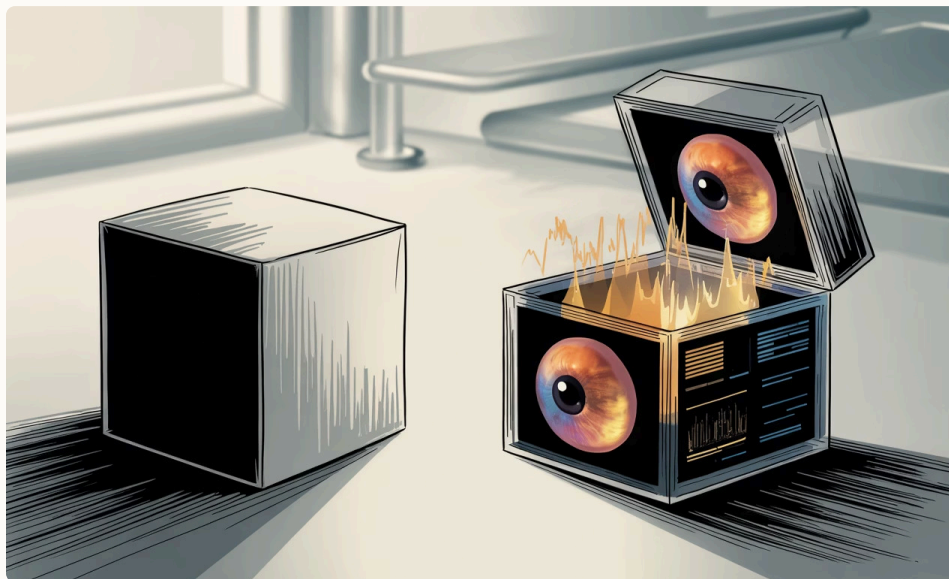
Too few ophthalmologists to screen everyone.

AI can analyze retinal images at scale, low cost.

But AI alone is not enough.



The Black Box Problem



- When an AI system says “severe diabetic retinopathy,” the doctor needs to know why.
- Clinical decisions cannot be made on blind faith.
- If AI cannot explain itself, the doctor cannot catch mistakes.

Without XAI

No reasoning. No evidence. No trust.



The doctor cannot verify the decision.

With XAI

Diagnosis + Reasoning. Transparent. Auditable.



The doctor can inspect, question, and trust the result.



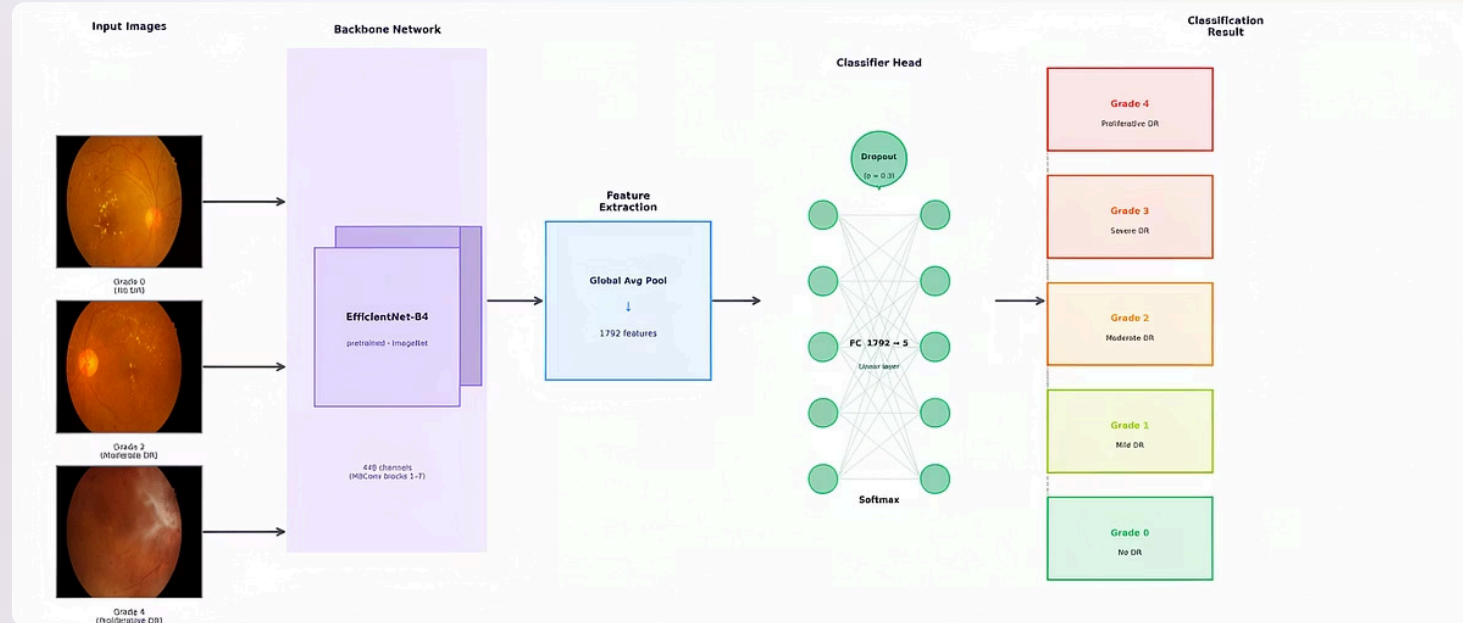
METHODOLOGY

Dataset

- EyePACS: 35,126 retinal images
- Dataset Imbalance: put weight on Grade 4
- Dataset Distribution: 75% for Training, 15% for Validation and 15% for Testing

Model Architecture

- **Backbone:** EfficientNet-B4 (ImageNet transfer learning)
- Global Average Pooling $\rightarrow$ 1,792 features
- Dropout ($p=0.3$) $\rightarrow$ Fully Connected (1,792 $\rightarrow$ 5)
- **Training:** Adam + weighted cross-entropy
- 10 epochs with cosine learning rate scheduler



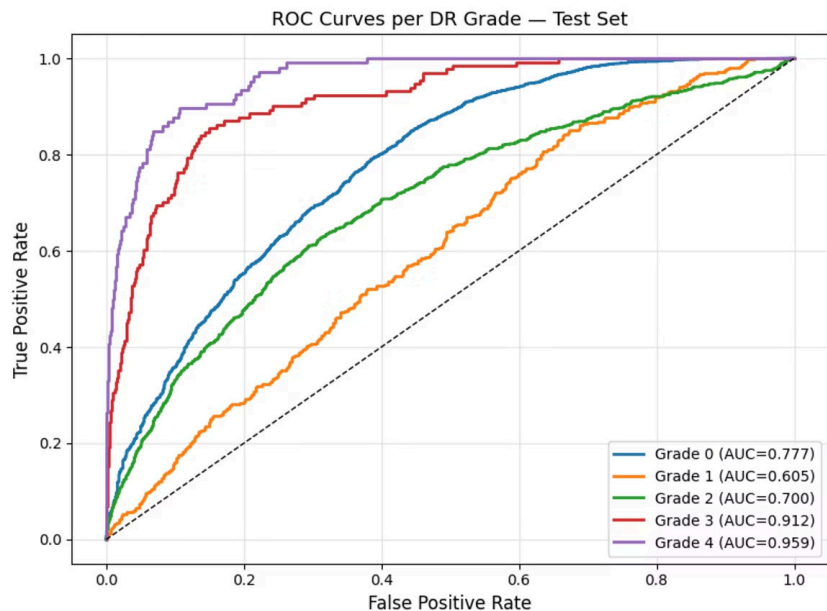
RESULTS

Model Performance

0.777

Macro AUROC

Across all 5 severity grades (one-vs-rest)



EXPLAINABILITY

What Does Explainable AI Actually Mean?

When the model makes a prediction, we ask it three questions.

1

Where did you look?

2

What did you find?

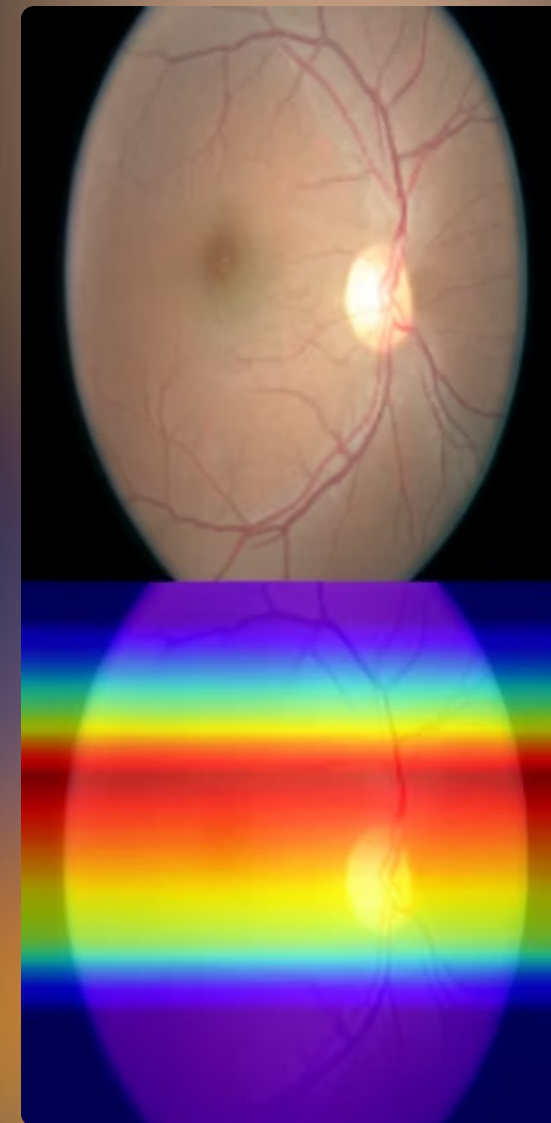
3

Why did that lead to
this conclusion?

XAI gives us the tools to answer.

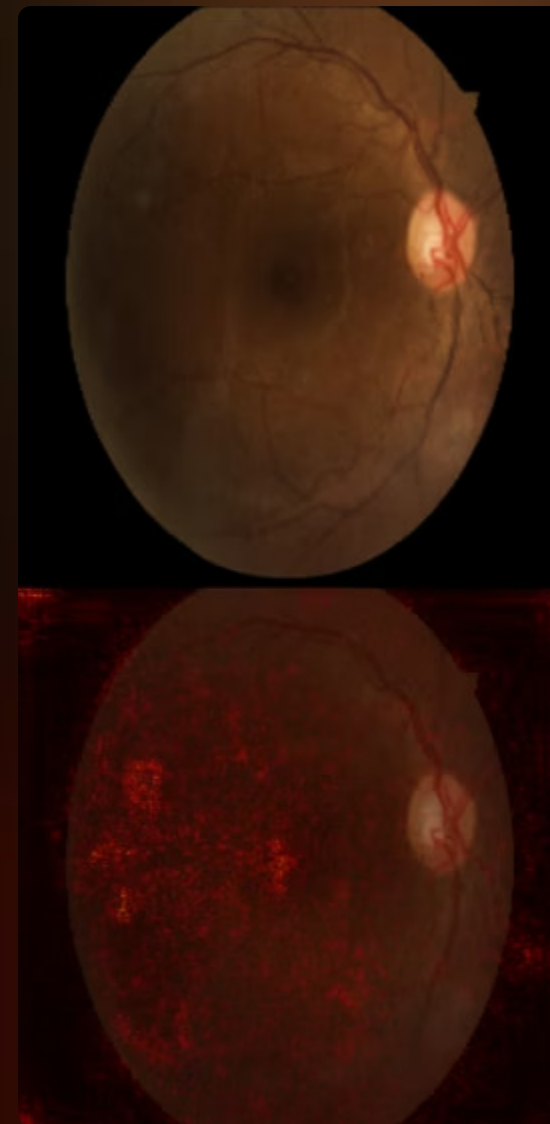
Grad-CAM: Visual Attention Maps

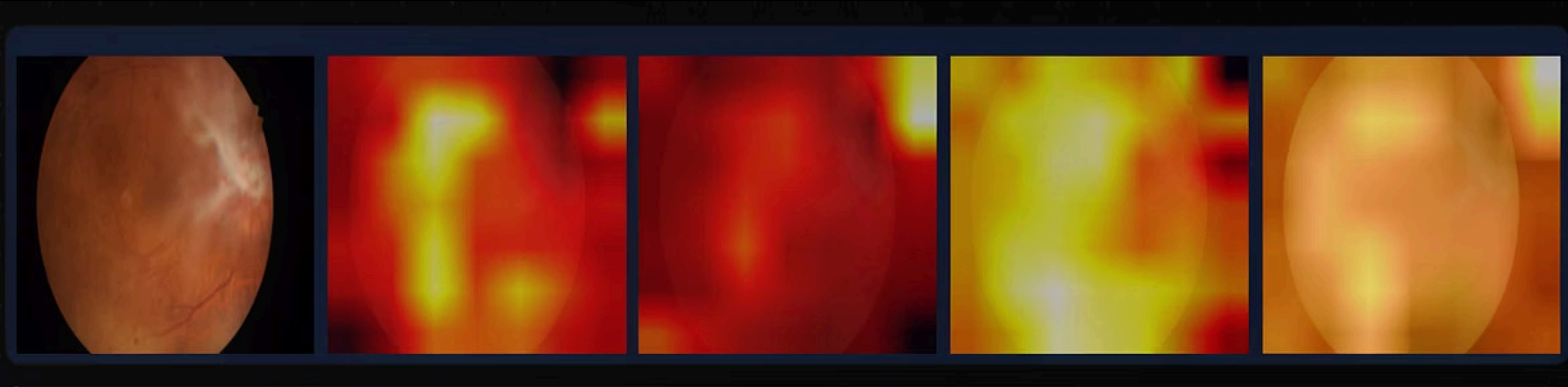
- Provides a visual explanation with a heatmap overlay directly on the retinal image
- Highlights the specific regions the model focused on when making its prediction
- **Red and orange areas** indicate where the model paid the most attention
- **blue areas** represent regions that were mostly ignored during the decision-making process



Integrated Gradients: Pixel Attribution

- Starts from a baseline blank image and gradually interpolates back to the original retinal image
- Computes gradients at each step along that path to see how the prediction changes
- Accumulates those gradients to measure each pixel's contribution to the final prediction
- Useful because it is mathematically rigorous, satisfies the completeness axiom, and reveals fine-grained pixel-level attribution





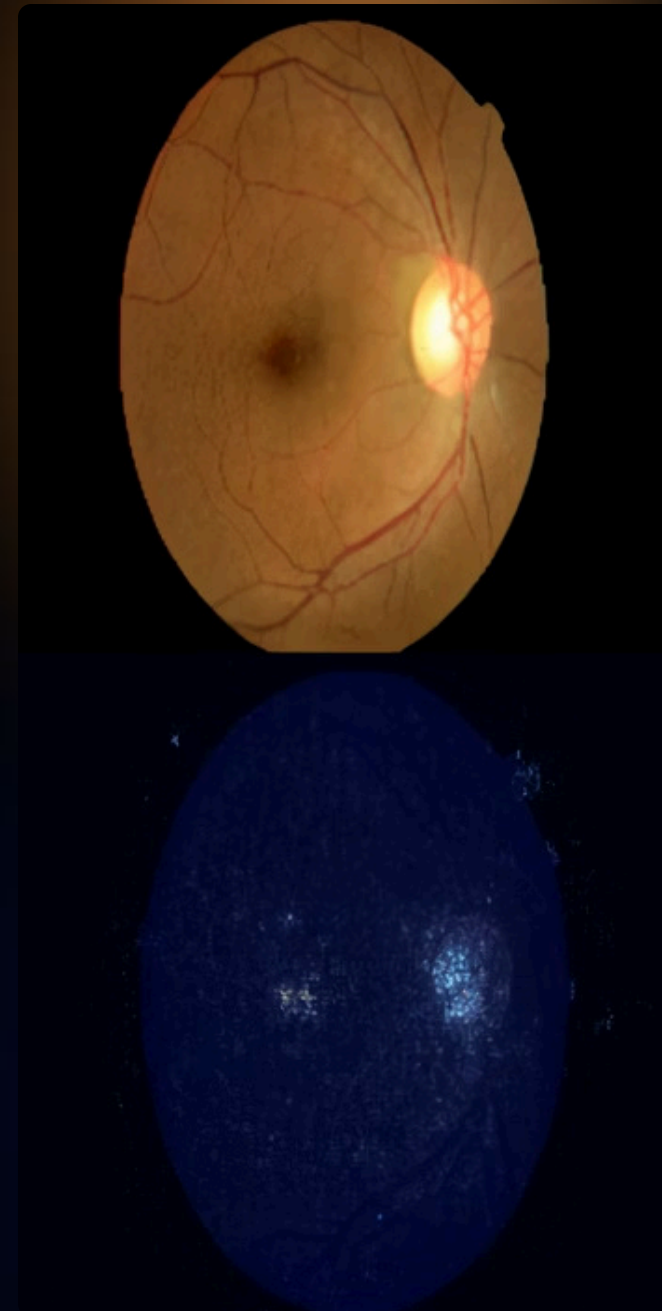
XAI METHOD DEEP DIVE

Feature Map Attention: Internal Filters

- In EfficientNet-B4, convolutional **internal filters** learn to detect specific visual patterns such as edges, textures, hemorrhages, and exudates
- Feature map attention shows which of those filters **activated most strongly** for a given input image
- This reveals what the model “sees” at an intermediate layer level, not just the final output prediction
- That makes it useful for understanding model behavior more deeply and for explaining why the network responds the way it does

SHAP: Pixel Importance Scores

- **SHAP** (**SHapley Additive exPlanations**) assigns an importance score to each pixel using **game theory** and **Shapley values**
- It answers a counterfactual question: if a pixel were removed or masked, how much would the prediction change?
- **Positive scores** push the prediction toward the predicted class, while **negative scores** push against it
- This is powerful for clinical interpretability because it gives a **quantitative, pixel-level breakdown** of what drove the diagnosis



Live Demonstration

Watch the AI explain its own decision in real time.

CONCLUSION

What We Have Demonstrated

Future Work

- Add LIME for region-level explanations
- Run user study with clinical ophthalmologists to evaluate clinical utility
- Fine-tune on Grade 2 vs Grade 4 boundary cases using misclassification analysis

The background features a light gray circular gauge with tick marks and a needle, overlaid with a faint, glowing tree-like structure. A small key icon is visible on the right side. Two horizontal lines frame the central text.

Thank You